Electrical & Electrician Formula Reference

FIRST-PRINCIPLES ELECTRICS → AC / THREE-PHASE → TESTING, TRANSFORMERS, MOTORS, INDUSTRIAL CALCULATIONS

Greek used below: ρ rho · φ phi · η eta · ω omega · τ tau

1 FUNDAMENTAL QUANTITIES

CURRENT

$I = Q / t$

I equals Q divided by t.

$Q = I \times t$

Q equals I times t.

I=current (A) · Q=charge (C) · t=time (s) · 1 A = 1 C/s

VOLTAGE

$V = E / Q$

V equals E divided by Q.

$E = V \times Q$

E equals V times Q.

V=volts · E=energy (J) · Q=coulombs · 1 V = 1 J/C

Voltage = energy per coulomb across a load. Loads transfer energy, not charge.

POWER & ENERGY

$P = E / t$

P equals E divided by t.

$E = P \times t$

E equals P times t.

1 W = 1 J/s

$P = V \times I$

P equals V times I.

$I = P/V \quad \cdot \quad V = P/I$

Rearranged forms of P = V × I.

2 OHM'S LAW

$V = I \times R$

V equals I times R.

$I = V / R$

I equals V divided by R.

$R = V / I$

R equals V divided by I.

V=volts · I=amperes · R=ohms (Ω)

3 RESISTIVE POWER / HEATING

$P = I^2 \times R$

P equals I squared times R.

$P = V^2 / R$

P equals V squared divided by R.

Also P = V × I — see §1.

Cable heating rises with the square of current.

4 SERIES CIRCUITS

RESISTANCE

$R_{total} = R1+R2+R3+…$

Total R equals R one plus R two, and so on.

VOLTAGE

$V_{total} = V1+V2+V3+…$

Total V equals V one plus V two, and so on.

CURRENT

$I1 = I2 = I3 = I_{total}$

Current is the same through every series component.

VOLTAGE DIVIDER

$V_x = V_s \times (R_x/R_{total})$

V x equals supply V times R x over total R.

Series: same current, voltage divides.

5 PARALLEL CIRCUITS

VOLTAGE

$V1 = V2 = V3 = V_{supply}$

Each branch voltage equals the supply voltage.

CURRENT — KCL

$I_{total} = I1+I2+I3+…$

Total current equals the sum of branch currents.

RESISTANCE

$1/R_{total} = 1/R1+1/R2+1/R3+…$

Reciprocal of total R equals sum of reciprocals.

$R_{total} = (R1 \times R2)/(R1+R2)$

Two resistors: product over sum.

Parallel: same voltage, current divides.

6 KIRCHHOFF'S LAWS

CURRENT LAW — KCL

$\Sigma I_{in} = \Sigma I_{out}$

Sum of currents in equals sum of currents out.

VOLTAGE LAW — KVL

$\Sigma V = 0$

Sum of voltages round a closed loop is zero.

$V_{supply} = V_{drop1}+V_{drop2}+…$

Practical form of KVL.

7 CABLE RESISTANCE

$R = \rho \times L / A$

R equals rho times L divided by A.

R=resistance · ρ (rho)=resistivity · L=length · A=cross-sec. area

$R_{loop} = 2 \times \rho \times L / A$

Loop R equals two times rho times L over A.

Use factor 2 only when L is the one-way length.

Longer cable → more R. Larger area → less R.

8 VOLTAGE DROP & CABLE LOSS

$\Delta V = I \times R$

Voltage drop equals current times resistance.

$VD\% = (\Delta V/V_{supply}) \times 100$

VD % equals ΔV over supply V, times 100.

$\Delta V = (mV/A/m \times I \times L)/1000$

mV per A per m, times I, times L, over 1000.

mV/A/m value must come from the applicable cable/design tables.

$P_{loss} = I^2 \times R$

Power loss equals current squared times R.

9 ENERGY / BILLING

$Energy\ (kWh) = P\ (kW) \times t\ (h)$

Energy equals power times time.

$Cost = kWh \times tariff$

Cost equals kilowatt-hours times tariff.

1000 W = 1 kW · 1000 Wh = 1 kWh

10 EFFICIENCY

$\eta = P_{out} / P_{in}$

Eta equals output power over input power.

$\eta\% = (P_{out}/P_{in}) \times 100$

Efficiency % equals that ratio times 100.

η (eta) = efficiency

11 AC FUNDAMENTALS

$f = 1/T \quad \cdot \quad T = 1/f$

Frequency and period are reciprocals.

f=frequency (Hz) · T=period (s)

50 Hz → T = 0.02 s = 20 ms

$\omega = 2\pi f$

Omega equals two pi times frequency.

ω (omega) = angular frequency

$v(t) = V_{peak} \times \sin(2\pi ft)$

Instantaneous voltage — sine-wave reference.

12 RMS & PEAK AC VALUES

$V_{rms} = V_{peak} / \sqrt{2}$

RMS V equals peak V over root two.

$V_{peak} = V_{rms} \times \sqrt{2}$

Peak V equals RMS V times root two.

$I_{rms} = I_{peak} / \sqrt{2} \quad \cdot \quad I_{peak} = I_{rms} \times \sqrt{2}$

Same relationship for current.

230 V RMS ≈ 325 V peak

13 CAPACITORS

$Q = C \times V$

Charge equals capacitance times voltage.

C=capacitance (F)

$E = \frac{1}{2} C V^2$

Energy equals half C times V squared.

$X_C = 1 / (2\pi fC)$

X C equals one over two pi f C.

Higher frequency → lower capacitive reactance.

14 INDUCTORS

$V = L \times (\Delta I/\Delta t)$

Voltage equals L times change in I over change in t.

$E = \frac{1}{2} L I^2$

Energy equals half L times I squared.

$X_L = 2\pi fL$

X L equals two pi f L.

Higher frequency → higher inductive reactance.

15 IMPEDANCE

$Z = V/I \quad \cdot \quad I = V/Z$

Impedance equals voltage over current.

$Z = \sqrt{R^2 + (X_L-X_C)^2}$

Series R-L-C impedance.

Z=ohms · R=resistance · XL=inductive react. · XC=capacitive react.

16 AC POWER

$S = V \times I$

Apparent power (VA).

$P = V \times I \times \cos\varphi$

Real power (W).

$Q_{reactive} = V \times I \times \sin\varphi$

Reactive power (var).

$S^2 = P^2 + Q^2$

Power triangle.

$PF = \cos\varphi = P/S$

Power factor.

φ (phi) = phase angle between V and I

Purely resistive load: PF ≈ 1, so P ≈ V × I.

17 TRANSFORMERS

$V_p/V_s = N_p/N_s$

Voltage ratio equals turns ratio.

$I_p/I_s = N_s/N_p$

Current ratio is the inverse turns ratio.

$V_p \times I_p \approx V_s \times I_s$

Ideal power in ≈ power out.

p=primary · s=secondary · N=number of turns

18 SINGLE-PHASE POWER

$P = V \times I$

Resistive load.

$P = V \times I \times PF$

General AC load.

19 THREE-PHASE POWER

$P = \sqrt{3} \times V_L \times I_L \times \cos\varphi$

Real power, balanced 3-phase.

$S = \sqrt{3} \times V_L \times I_L$

Apparent power.

$Q = \sqrt{3} \times V_L \times I_L \times \sin\varphi$

Reactive power.

VL=line-to-line V · IL=line current

EU 230/400 V system: Line–Neutral ≈ 230 V · Line–Line ≈ 400 V

20 STAR & DELTA

STAR / WYE

$V_L = \sqrt{3} \times V_{phase} \quad \cdot \quad I_L = I_{phase}$

Line V is √3 × phase V; line I = phase I.

DELTA

$V_L = V_{phase} \quad \cdot \quad I_L = \sqrt{3} \times I_{phase}$

Line V = phase V; line I is √3 × phase I.

21 RCD / RESIDUAL CURRENT

$I_\Delta = |I_L - I_N|$

Residual current = |Live I − Neutral I|.

IΔ=residual/differential current

e.g. 5.000 A − 4.975 A = 0.025 A = 25 mA

MCB → excess current. RCD → current imbalance.

22 FAULT CURRENT / EARTH-FAULT LOOP

$I_{fault} = U_0 / Z_s$

Fault current equals U zero over Zs.

$Z_s = Z_e + (R1+R2)$

Zs equals Ze plus R one plus R two.

$Z_s \le U_0 / I_a$

Zs must not exceed U zero over Ia.

U0=nom. V to earth · Ze=external loop Z · R1=line R · R2=CPC R · Ia=device trip current

Actual permitted Zs & disconnection times: use applicable regs/device tables.

23 Basic Circuit Design Relationship

$$I_b \le I_n \le I_z$$

I b is less than or equal to I n, less than or equal to I z.

Ib=design/load current · In=device rating · Iz=cable capacity — **load → protective device → cable**

24 PROTECTIVE CONDUCTOR (ADIABATIC)

ADVANCED

$S = \sqrt{I^2 t} / k$

S equals root of I-squared-t, over k.

$S = I\sqrt{t} / k$

Equivalent form.

S=required c.s.a. · I=fault current · t=disconnection time · k=material/insulation constant. Use with applicable regs/tables.

25 MOTOR BASICS

$n_s = 120f / p$

Sync. speed = 120 × f, over pole count.

ns=sync. speed (rpm) · p=number of poles

$Slip\% = ((n_s-n_r)/n_s) \times 100$

Slip % = (sync − rotor speed)/sync, ×100.

nr=actual rotor speed

$P_{mech} = \tau \times \omega$

Mechanical power = torque × angular velocity.

τ (tau) = torque (N·m) · ω (omega) = ang. velocity (rad/s)

26 PREFIXES & CONVERSIONS

1 kA = 1000 A · 1 A = 1000 mA · 1 mA = 0.001 A

1 kV = 1000 V · 1 kW = 1000 W · 1 kΩ = 1000 Ω

1 MΩ = 1,000,000 Ω · 1 mF = 10^{-3} F · 1 µF = 10^{-6} F · 1 nF = 10^{-9} F

√2 ≈ 1.414 · √3 ≈ 1.732 · π ≈ 3.142

MENTAL MODEL — HOLD ONTO THESE

Current: charge per second.

Voltage: potential-energy diff. per coulomb.

Resistance: opposition to current.

Power: rate of energy transfer.

Load: transfers energy, not charge.

Series: same current, V divides.

Parallel: same voltage, I divides.

MCB: protects against overcurrent.

RCD: detects L–N current imbalance.

PE: protective fault-current path.

Neutral: near earth potential, not "used-up Live".

Ib≤In≤Iz: load → device → cable.